# Indices, Surds and Polynomials

## Section 1: Fractional Indices and Exponential Equations

**1** Evaluate the exact numerical value of $16^{-\frac{3}{4}} \times 27^{\frac{2}{3}}$.

**a.** $\frac{9}{8}$

**b.** $\frac{8}{9}$

**c.** $\frac{3}{4}$

**d.** $\frac{27}{16}$

**2** Solve the exponential equation $2^{2x+1} - 9(2^x) + 4 = 0$ for all real $x$.

**a.** $x = -1$ and $x = 2$

**b.** $x = 1$ and $x = 4$

**c.** $x = 0$ and $x = 2$

**d.** $x = -2$ and $x = 1$

**3** Simplify the index expression $\frac{(4x^3y^{-2})^{\frac{1}{2}}}{2x^{-1}y^3}$.

**a.** $x^{\frac{5}{2}}y^{-4}$

**b.** $2x^{\frac{5}{2}}y^{-4}$

**c.** $x^2y^{-4}$

**d.** $x^{\frac{5}{2}}y^2$

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## Section 2: Surds and Rationalising Denominators

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**4** Rationalise the denominator and simplify $\frac{4+\sqrt{3}}{2-\sqrt{3}}$ into the form $a+b\sqrt{3}$.

**a.** $11+6\sqrt{3}$

**b.** $5+6\sqrt{3}$

**c.** $11-6\sqrt{3}$

**d.** $7+4\sqrt{3}$

**5** Express $(3-2\sqrt{5})^2$ in the simplified form $p+q\sqrt{5}$.

**a.** $29-12\sqrt{5}$

**b.** $29+12\sqrt{5}$

**c.** $11-12\sqrt{5}$

**d.** $19-6\sqrt{5}$

**6** Solve the radical equation $\sqrt{2x+7}-x=2$ for all valid real solutions.

**a.** $x=1$ only

**b.** $x=1$ and $x=-3$

**c.** $x=3$ only

**d.** No real solutions

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## Section 3: Remainder and Factor Theorems

**7** When $P(x)=2x^3+ax^2-5x+6$ is divided by $(x-2)$, the remainder is $16$. Find the value of $a$.

**a.** $a=1$

**b.** $a=-2$

**c.** $a=3$

**d.** $a=4$

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**8** Given that $(2x - 1)$ is a factor of $f(x) = 2x^3 + 3x^2 - 8x + c$, find the constant $c$.

**a.** $c = 3$

**b.** $c = -3$

**c.** $c = 6$

**d.** $c = -2$

**9** Factorise completely the cubic polynomial $g(x) = x^3 - 4x^2 - 7x + 10$.

**a.** $(x - 1)(x + 2)(x - 5)$

**b.** $(x + 1)(x - 2)(x + 5)$

**c.** $(x - 1)(x - 2)(x - 5)$

**d.** $(x + 1)(x + 2)(x - 5)$

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## Section 4: The Binomial Theorem

**10** Find the coefficient of $x^3$ in the binomial expansion of $(1 + 2x)^6$.

**a.** 160

**b.** 80

**c.** 240

**d.** 120

**11** Find the term independent of $x$ (the constant term) in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

**a.** 240

**b.** 160

**c.** 60

**d.** 192

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**12** In the expansion of $(1+ax)^5(1-x)$, the coefficient of $x$ is $9$. Find the value of $a$.

**a.** $a=2$

**b.** $a=1$

**c.** $a=3$

**d.** $a=-2$

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